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Flexible display panels

Abstract

According to an example, a computing device comprises a base portion and a lid portion rotatably coupled to the base portion. The lid portion comprises a platform, an aperture to receive the platform, and a flexible display panel disposed on the platform. Upon a deployment of the platform, an area of the flexible display panel on an upper side of lid portion is to increase such that the area overlaps with the base portion.

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Background/Summary

BACKGROUND

(1) Electronic devices such as computing devices may comprise flexible display panels to display image contents. A flexible display panel may be bent, rolled, or curved without damaging the integrity of the computing device. Flexible display panels may be based on flexible substrates such as plastic, metal, or flexible glass.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) Features of the present disclosure are illustrated by way of example and are not limited in the following figure(s), in which like numerals indicate like elements, in which:
- (2) FIG. 1A shows a detailed view of a computing device including a platform in a retracted position, according to an example of the present disclosure;
- (3) FIG. 1B shows a detailed view of the computing device of FIG. 1A with the platform in a deployed position;
- (4) FIG. 2A shows a detailed view of a computing device with a platform movable in a longitudinal direction, according to an example of the present disclosure;
- (5) FIG. 2B shows a detailed view of the computing device of FIG. 2A upon deploying of the platform;
- (6) FIG. 3A shows a detailed view of a computing device comprising a lid portion including a flexible display panel, according to an example of the present disclosure;
- (7) FIG. 3B shows the computing device of FIG. 3A upon a deployment of a platform of the lid portion;
- (8) FIG. 4A shows a detailed view of a computing device comprising a protruding element in a first position, according to an example of the present disclosure;
- (9) FIG. 4B shows the computing device of FIG. 4A with the protruding element in a second position;
- (10) FIG. 5A shows a side view of a computing device comprising a platform in a retracted position, according to an example of the present disclosure;
- (11) FIG. 5B shows a side view of a computing device comprising a deployment member located in a lid portion, according to an example of the present disclosure;
- (12) FIG. 5C shows a side view of a computing device comprising a platform in a deployed position, according to an example of the present disclosure;
- (13) FIG. 6A shows a side view of a computing device comprising a lid portion substantially perpendicular to a base portion, according to an example of the present disclosure;
- (14) FIG. 6B shows a side view of a computing device having a platform at an angle with respect to a lid portion, according to an example of the present disclosure;
- (15) FIG. 7A shows a detailed view of a computing device comprising a stand feature aligned with a surface of the base portion, according to an example of the present disclosure;

(16) FIG. 7B shows the computing device of FIG. 7A upon the stand feature protrudes from the base portion;

(17) FIG. 8 shows a method for deploying a platform of a lid portion of a computing device, according to an example of the present disclosure;

(18) FIG. 9 shows a method for unlocking a movement of a platform of a computing device upon an angle between the lid portion and the base portion is greater than a threshold angle, according to an example of the present disclosure.

DETAILED DESCRIPTION

(19) For simplicity and illustrative purposes, the present disclosure is described by referring mainly to examples. In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. It will be readily apparent, however, that the present disclosure may be practiced without limitation to these specific details. In other instances, some methods and structures have not been described in detail so as not to unnecessarily obscure the present disclosure.

(20) Throughout the present disclosure, the terms “a” and “an” are intended to denote at least one of a particular element. As used herein, the term “includes” means includes but not limited to, the term “including” means including but not limited to.

(21) Electronic devices such as computing devices may use flexible display panels to display image content. Such flexible display panels comprise a plurality of pixels formed on a thin and flexible substrate, where the flexible substrate comprises materials such as plastic, metal, or flexible glass. In this fashion, even when the flexible display panel may be bent, folded, or rolled up, the flexible display panel will be capable of displaying the desired image content without suffering any damage. Examples of such flexible display panels include flexible liquid crystal display panels, flexible plasma display panels, flexible e-ink display panels, flexible organic light-emitting diode panels, flexible electro-wetting display panels, among others.

(22) In order to transmit the image content to the flexible display panel, computing devices may send data associated with the image content to a controller operatively connected to the flexible display panel. When having a flexible touch display panel, the controller may issue a signal associated with a contact on a location of the flexible touch display panel.

(23) Among the alternative uses available for a flexible display panel, a computing device may include the flexible display panel in order to enable multiple display sizes at the discretion of the user. Therefore, the same computing device may have a range of display sizes instead of a screen with fixed screen size. To re-size an effective display panel size, the computing device may have to perform some retraction and deployment operations in order to increase/decrease the effective display panel size on which image content is to be displayed. In some examples, some of these operations may imply using additional tools to remove some fixed elements in order to re-size the flexible display panel. For instance, when the computing device is a laptop comprising a lid portion rotatably coupled to a base portion, the flexible display panel configuration may be fixed via a series of screws. In order to expand/reduce the effective display panel size, a user may have to use tools to remove the screws, then, the user may have to set the desired effective display panel size, and then, put the screws back on the lid portion to ensure that the integrity of the computing device is not affected by the re-sizing process.

(24) Throughout the present disclosure, the term “effective display panel size” will be used to refer to a portion of the flexible display panel on which an image content will be displayed for a user, i.e., the flexible display panel dimensions on which users will see the image content.

(25) Disclosed herein are examples of computing devices and methods to modify an effective display panel size without using additional tools. Hence, different examples of computing devices, and methods are described.

(26) Referring now to FIG. 1A, a computing device **100** comprising a platform **121** in a retracted position is shown. The computing device **100** comprises a lid portion **120** rotatably coupled to a

base portion **110** via a hinge **130**. The lid portion **110** comprises a flexible display panel **123** disposed on a platform **121** (represented in dashed lines). The hinge **130** enables a rotation of the lid portion **120** with respect to the base portion **110** of the computing device **100**, where the base portion **110** comprises a keyboard **111** on an upper surface.

(27) In FIG. **1A**, the platform **121** is in a retracted position within an aperture of the lid portion **120** and the flexible display panel **123** is disposed on the platform **121**. In the retracted position, the platform **121** does not extend away from a bottom side **120B** of the lid portion **120**, and therefore, the platform **121** is in the lid portion **120**.

(28) As previously explained, flexible display panels may be used to enable a range of effective display panel sizes without using additional tools. In FIG. **1A**, the flexible display panel **123** covers the platform **121** of the lid portion **120** such that, if the platform **121** is deployed from the lid portion **120** through the aperture on the bottom side **120B** of the lid portion **120**, an area of flexible display panel **123** on an upper side of the lid portion **120** is to increase.

(29) Referring now to FIG. **1B**, the computing device **100** of FIG. **1A** with the platform **121** in a deployed position is shown. In the deployed position, the platform **121** of the computing device **100** fully extends from the bottom side **120B** of the lid portion **120**. As a result of the deployment in the transversal direction, the flexible display panel **123** extends over the base portion **110** such that the base portion **110** and the flexible display panel **123** overlap. In other examples, the aperture may be positioned at a different position instead of the bottom side **120B** of the lid portion **120**. In an example, the aperture may be provided on a top side of the lid portion **120**.

(30) In order to deploy the platform **121** of the computing device **100**, an upper surface of the lid portion **120** and an upper surface of the base portion **110** are positioned at an angle equal to 180 degrees or greater than 180 degrees. In the deployed position, the platform **121** and the flexible display panel **123** cover the keyboard **111** of the base portion **110**. However, when having the aperture of the lid portion **120** on a different side of the lid portion **120**, the platform **121** may deploy in a different direction instead of the transversal direction towards the base portion **110**. In an example, when the aperture is on the top side of the lid portion **120**, the platform **121** may deploy in a transversal direction away from the base portion **110**.

(31) Throughout the description, the term “retracted position” will be used to refer to a position of the platform in which the platform is fully contained in the lid portion of the computing device. In addition, the term “deployed position” will be used to refer to a position of the platform in which the platform fully extends away from the aperture of the lid portion.

(32) Referring now to FIG. **2A**, a computing device **200** having a platform **221** movable in a longitudinal direction is shown. The computing device **200** comprises a lid portion **220** rotatably attached to a base portion **210** of the computing device **200** via a hinge **230**. The lid portion **220** comprises a flexible display panel **223** disposed on a platform **221** (represented in dashed lines). As previously explained in reference to FIGS. **1A** and **1B**, the platform **221** is movable within an aperture of the lid portion **220**. In FIG. **2A**, the aperture is provided on a lateral side **220C** of the lid and the platform **121** is in a retracted position. In the retracted position, the platform **221** does not protrude from the lateral side **220C** of the lid portion **220** of the computing device **200**.

(33) In the computing device **200**, the lid portion **220** further comprises a removable peripheral **201** attached on a top side of the lid portion **220**. Due to the peripheral **201** can be removed, users may remove the peripheral **201** and re-locate it in alternative locations. In an example, the peripheral **201** may be a webcam. In other examples, the peripheral **201** may be a speaker or a microphone.

(34) Due to the aperture of the lid portion **220** of the computing device **200** is on a lateral side **220C** of the lid portion **220** instead of the bottom side of the lid portion **220** (bottom side **120B** in FIGS. **1A** and **1B**), a deployment of the platform **221** will result in a movement parallel to a rotation axis defined by the hinge **230**. Therefore, although the lid portion **220** and the base portion **210** may be at an angle lower than 180 degrees, the deployment of the platform **221** will not result in a contact between the base portion **210** and the platform **221**.

(35) Referring now to FIG. 2B, the computing device **200** upon a deployment of the platform **221** is shown. Due to the aperture is provided on the lateral side **220C** of the lid portion **220**, the deployment is performed in a longitudinal direction. In the computing device **200** of FIG. 2B, the platform **221** does not fully protrude from the aperture of the lid portion **220**. Instead, the platform **221** partially protrudes from the aperture of the lid portion **220**. Hence, the position of the platform corresponds with an intermediate position between the retracted position (shown in FIG. 2A) and the deployed position (in which the platform **221** fully protrudes from the aperture).

(36) In FIG. 2B, an area of the flexible display panel **223** on the upper side of the lid portion **220** is greater than the area of flexible display panel **223** on the upper side of the lid portion before the deployment. In other words, the effective display panel size of the computing device **200** of FIG. 2B is greater than the effective display panel size of the computing **200** of FIG. 2A.

(37) As previously explained, the peripheral **201** can be positioned at different locations. In FIG. 2B, the peripheral **201** has been moved to remain centered with respect to the effective display panel size of the computing device **200**. Similarly, the keyboard **211** has been removed from the base portion **210** to be used separately in order to enhance the user experience.

(38) Referring now to FIG. 3A, a detailed view of a computing device **300** comprising a lid portion **320** rotatably coupled to a base portion **310** via a hinge **330** is shown. The base portion **310** includes a keyboard. In an example, the keyboard may be removable. In other examples, the base portion **310** may include additional elements such as a touchpad.

(39) The lid portion **320** of the computing device **300** comprises a platform **321** (represented in dashed lines in FIG. 3A) in a retracted position, an aperture **322** to receive the platform **321** on a bottom side **320B** of the lid portion **320**, and a flexible display panel **323** disposed on the platform **321**. In the example of FIG. 3A, an upper side of the platform **321** is covered by the flexible display panel **323** such that the effective display panel size corresponds with an area on an upper side of the lid portion **320**. To enable a range of effective display panel sizes, the flexible display panel **323** also covers a lower side of the platform **321** such that a movement of the platform **321** from the retracted position towards a deployed position results in a modification of the effective display panel size. In FIG. 3A, the flexible display panel **323** is bent around an edge of the platform **321** located next to the hinge **130**. Thus, a movement of the platform **321** away from the lid portion **320** will result in an increase of the effective display panel size, i.e., an area on which users will see the image content on the flexible display panel **323**.

(40) Referring now to FIG. 3B, the computing device **300** having the platform **321** in the deployed position is shown. As a result of the deployment, an area of the flexible display panel **323** on an upper side of the lid portion **320** is to increase such that the area overlaps with the base portion **310**.

(41) In the computing device **300**, the deployment of the platform **321** results in a decrease of the area of flexible display panel **323** on a lower side of the platform **321**. As explained in reference to FIG. 3A, to enable an increase of the effective display panel size based on a position of the platform **321** with respect to the aperture **322**, the flexible display panel **323** covers the upper side of the lid portion **320** and a lower side of the platform **321**. In FIG. 3B, a first edge of the flexible display panel **123** on the upper side of the lid portion **320** is attached to a top side **320A** of the lid portion **320**, the flexible display panel **323** and the lower side of the platform **321** overlap in a lower side overlap area, and, upon the deployment of the platform **321**, the lower side overlap area is to decrease with respect to the lower side overlap area on the lower side of the platform **321** before the deployment of the platform **321**.

(42) In some examples, the deployment of the platform **321** is in response to a trigger event. Such trigger event may be a signal received by a controller of the computing device **300** (for instance, a trigger signal associated with a deployment of the platform **321**), a relative position between the lid portion **320** and the base portion **310** (for instance, exceeding a threshold angle between the lid portion **320** and the base portion **310**), actuating an actuator member of the computing device (for instance, actuating a deployment member located in the lid portion **320**), among others.

(43) According to some examples, the lid portion of the computing device may rotate with respect to the base portion of the computing device in a range from 0 degrees to substantially 360 degrees. In other examples, the lid portion may rotate with respect to the base portion in a range from 0 degrees to substantially 180 degrees. In some examples, the computing device may comprise a locking member to lock a position of the platform with respect to the lid portion based on a relative position between the lid portion and the base portion. For instance, referring to the computing device **300** of FIGS. 3A and 3B, the locking member may unlock the movement of the platform **321** upon the lid portion **320** and the base portion **310** are at an angle equal to or greater than 180 degrees. In this fashion, a contact between the platform **321** and the base portion **310** is prevented. In an example, the movement of the platform **321** may be unlocked by the locking member based on a signal of a sensor of the computing device **300**. However, in other examples, the movement of the platform **321** may be unlocked by the locking member based on mechanical elements, such as a self-locking cam to unlock the platform **321** upon the lid portion **320** and the base portion **310** are at an angle greater than 180 degrees.

(44) Referring now to FIG. 4A, a computing device **400** comprising a protruding element **425** in a first position is shown. The computing device **400** comprises a lid portion **420** rotatably coupled to a base portion **410** via a hinge **430**, where the lid portion **420** comprises a platform **421** (represented in dashed lines), an aperture **422** on a bottom side **420B** of the lid portion **420** to receive the platform **421**, a flexible display panel **423** to cover an upper side of the lid portion **420** and a lower side of the platform **421**, a track **424** along a lateral side **420D** of the lid portion **420**, and the protruding element **425** on a lateral side of the platform **421**. The protruding element **425** protrudes from the lateral side **420D** of the lid portion **420** and is movable along the track **424** between a first position and a second position. The first position corresponds to a position in which the platform **421** does not protrude from the lid portion **420**. The second position corresponds to a position in which the platform is fully deployed from the lid portion **220**. In the computing device **200A**, the protruding element **225** is in the first position. In an example, the first position of the protruding element **425** corresponds to the retracted position of the platform **421** and the second position corresponds to the deployed position of the platform **421**.

(45) In FIG. 4A, a first end of the flexible display panel **423** located on the upper side of the lid portion **420** is attached to the lid portion **420**. In particular, the first end of the flexible display panel **423** is attached to a top side **420A** of the lid portion **420**. As previously explained in FIGS. 3A and 3B, the flexible display panel **423** also covers the lower side of the platform **421** such that the deployment of the platform **421** results in an increase of the portion of flexible display panel **423** on the upper side of the lid portion **420**, i.e., the effective display panel size.

(46) In some examples, a movement of the platform **421** with respect to the lid portion **420** may be locked by a locking member. Thus, although a user aims to deploy the platform **421**, the locking member prevents the platform **421** from extending away from the aperture **422**. To unlock the movement of the platform **421**, the locking member may have to receive an unlocking signal from the controller. In other examples, the locking member may comprise mechanical elements to determine whether or not a current configuration of the computing device **400** (for instance, an angle between the lid portion **420** and the base portion **410**) meets the deployment criteria. In an example, the deployment criteria to unlock the movement of the platform **421** may be if an angle between the lid portion **420** and the base portion **410** is within a threshold angle range (for instance, a range from 180 degrees to 270 degrees). Thus, if the lid portion **420** and the base portion **410** are at an angle outside the threshold angle range, the deployment of the platform **421** is locked by the locking member.

(47) Referring now to FIG. 4B, the computing device **400** is shown when the protruding element **425** is in a second position. In the second position, the platform **421** fully extends from the aperture **422** of the lid portion **420**. As indicated by the arrows, the movement of the protruding element **425** from the first position (represented in FIG. 4A) to the second position results in an increase of the

effective display panel size.

(48) In FIG. 4B, the lid portion **420** of the computing device **400** is at an angle wider than 180 degrees with respect to the base portion **410**. As previously explained, in some examples, the movement of the platform **421** may be locked by a locking member based on a relative angle between the lid portion **420** and the base portion **410**. For instance, the movement may be locked if the relative angle is outside a threshold angle range.

(49) Although the computing devices **300**, **400** of FIGS. 3A to 4B comprise the aperture **322**, **422** on the bottom side **320B**, **420B** of the lid portion **320**, **420**, it should be understood that in other examples alternative locations such as a lateral side (as previously explained in reference to FIGS. 2A and 2B) or the top side **320A**, **420A** may be possible. Based on the location of the aperture **322**, **422** of the lid portion **320**, **420**, the flexible display panel **323**, **423** may be fixedly attached to a different location instead of the top side **320A**, **420A** of the lid portion **320**, **420**.

(50) Referring now to FIG. 5A, a side view of a computing device **500A** comprising a platform **521** in a retracted position is shown. The computing device **500A** may correspond, for instance, with the computing devices **100**, **200**, **300** and **400** previously explained in reference to FIGS. 1A to 4B. Thus, the similarly-named elements of example computing devices may be similar in function and/or structure to the respective elements of other example computing devices, as they are described above. In FIG. 5A, the lid portion **520** comprises an aperture **522** to receive the platform **521** and the flexible display panel **523** is disposed on the platform **521**, where an upper side area **523A** represents an area of the flexible display panel **523** on an upper side of the lid portion **520** and a lower side area **523B** represents an area of the flexible display panel **523** on a lower side of the platform **521**.

(51) In the computing device **500A**, the base portion **510** comprises a sensor **526** to detect a position of the platform **521** with respect to the lid portion **520**. Based on the determined position, a deployment state of the platform **521** may be determined. In an example, the sensor **526** may determine a position of the platform **521** with respect to the retracted position. In some examples, the deployment state may be used to determine an image content on the flexible display panel **523**. In some other examples, the deployment state of the platform **521** may be associated with an interface region corresponding to the upper side area **523A**. In other examples, the interface region may correspond to the lower side area **523B**.

(52) In order to rotatably couple the base portion **510** and the lid portion **520**, the computing device **500A** comprises a hinge assembly **530**. In some examples, the hinge assembly **530** comprises a hinge and a locking member to lock a movement of the platform **521** with respect to the lid portion **520**. In response to an unlocking signal, the locking member is to unlock the movement of the platform **521**. In other examples, the hinge assembly **530** may comprise a hinge, a locking member to lock a movement of the platform with respect to the lid portion **520**, and a sensor to determine an angle between the lid portion **520** and the base portion **510**. Upon the sensor determines an angle within a threshold angle range, the locking member is to unlock the movement of the platform **521**.

(53) Although the sensor **526** of the computing device **500A** is in the base portion **510**, in other examples, alternative locations may be possible. In an example, the sensor **526** may be in the lid portion **520**, where the sensor is to determine a distance from the sensor **526** to the platform **521**. Based on the distance, the deployment state of the platform **521** may be determined.

(54) Referring now to FIG. 5B, a computing device **500B** comprising a deployment member **527** in the lid portion **520** is shown. The deployment member **527** is to move the platform **521** between a retracted position and a deployed position. In FIG. 5B, the deployment member **527** has moved the platform **521** such that the platform **521** extends away from the aperture of the lid portion **520**. Upon the deployment of the platform **521**, the upper side area **523A** of the flexible display panel **523** is to increase such that the upper side area **523A** overlaps with the base portion **510**. In FIG. 5B, an overlap area **524** represents a projection of the upper side area **523A** onto the base portion **510**. In addition, the deployment of the platform **521** results in a decrease of the lower side area

523B.

(55) In an example, the deployment member **527** is to deploy the platform **521** in response to a trigger event. The trigger event may comprise receiving a trigger signal associated with the deployment of the platform **521**, a trigger signal associated with an angular position of the lid portion **520** with respect to the base portion **510**, actuating an actuator member located in the lid portion **520**, among others. In some examples, the deployment member **527** may position the platform **521** in a set of positions between the retracted position of the platform **521** and the deployed position of the platform **521**. In an example, when the computing device **500B** corresponds with the computing device **400** of FIGS. **4A** and **4B**, the deployment member **327** may include a declutch state to enhance the user experience when users are moving the protruding element **425** along the track **424**. In the declutch state, the deployment member **427** does not exert forces towards the platform **421**.

(56) Referring now to FIG. **5C**, a computing device **500C** comprising the platform **521** in the deployed position is shown. As previously explained, the deployed position corresponds to a position in which the platform **521** is fully extended from the aperture of the lid portion **520**. In the deployed position, a projection of the upper side area **523A** onto the base portion **510** results in an overlap area **524**. As a result of the deployment, the flexible display panel **523** and a lower side of the platform **521** do not overlap. In other words, the flexible display panel **523** is on the upper side of the lid portion **520**. In some examples, in order to tighten the flexible display panel **523**, a distal edge of the platform **521** may comprise a torsion element coupled to an end of the flexible display panel **523**. As a result, the tensions along the length of the flexible display panel **523** will be maintained even when the relative position between the platform **521** and the lid portion **520** has changed.

(57) The computing device **500C** comprises a sensor **526A** in the base portion **510** to determine a position of a first reference element **526B** and a second reference element **526C**. In an example, based on the determined positions for the first reference element **526B** and for the second reference element **526C**, the sensor **526A** may determine the deployment state of the platform **521** and the angle between the lid portion **520** and the base portion **510**.

(58) In FIG. **5C**, the lid portion **520** is coupled to the base portion **510** via a hinge assembly **530** which enables angles wider than 180 degrees between the lid portion **520** and the base portion **510**. As previously explained, the hinge assembly **530** may comprise a locking member to lock a deployment of the platform **521**. In some examples, the locking member of the hinge assembly **530** may enable the deployment of the platform **521** upon the sensor **526A** determines an angle greater than 180 degrees between the lid portion **520** and the base portion **510**. However, in other examples, the hinge assembly **530** may unlock the deployment of the platform **521** upon an upper side of the lid portion **520** is substantially parallel to the upper side of the base portion **510**.

(59) Referring now to FIG. **6A**, a computing device **600A** comprising a lid portion **620** substantially perpendicular to a base portion **610** is shown. The lid portion **620** is rotatably coupled to the base portion **610** via a hinge mechanism **630** comprising a locking member and a sensor to determine an angle between an upper side of the lid portion **620** and an upper side of the base portion **610**. Upon determining that the angle is within a threshold angle range, the sensor may issue a signal to the locking member. In an example, in response to the signal, the locking member of the hinge assembly **630** unlocks a movement of the platform **621**. In other examples, the lid portion **620** may comprise a deployment member and the signal may also trigger the deployment of the platform **621**.

(60) In the example of FIG. **6A**, an end of the flexible display panel **623** is movably attached to a distal end **621A** of the platform **621**. However, in other examples, the flexible display panel **623** may be attached at a different location instead of the distal end **621A** of the platform **621**, such as a lower side of the platform **621**. In some examples, the flexible display panel **623** may be attached to the platform **621** via an elastic element to tighten the flexible display panel **623** on the upper side

of the lid portion **620**.

(61) Referring now to FIG. **6B**, a computing device **600B** with the platform **621** at an angle θ with respect the lid portion **620** is shown. In the example of FIG. **6B**, the platform **621** is in the deployed position. In the deployed position, the platform **621** is rotatable with respect to the lid portion **620** such that a curved display panel configuration is obtained. In order to enable the rotation of the platform **621** with respect to the lid portion **620**, the platform **621** comprises a platform hinge **628** on a proximal end of the platform **621**. Since the platform hinge **628** is located on the proximal end of the platform **621**, the rotation about the hinge **628** is unlocked upon the platform hinge **628** protrudes from the lid portion **620**. In an example, the platform hinge **628** protrudes from the lid portion **620** in the deployed position of the platform **621**. In some examples, the platform hinge **628** enables angles θ lower than 180 degrees. In some other examples, the angle θ may comprise values within an angle range from 160 degrees to 180 degrees.

(62) The platform **621** of the computing device **600B** further comprises a torsion element **629** on the distal end **621A** of the platform **621**, where the torsion element **629** is to tighten the flexible display panel **623** on the upper side of the lid portion **620**. In the example of FIG. **6B**, the torsion element **629** is fixedly attached to a second end of the flexible display panel **623**. However, in other examples, the torsion element **629** may at a different location such as the lower side of the platform **621** instead of the distal end **621A** of the platform **621**.

(63) In some examples, an intermediate region of the flexible display panel **623** may be fixedly attached to the lid portion **620** in order prevent the flexible display panel **423** from being loosen when the platform **621** is at the angle θ with respect the lid portion **620**.

(64) As previously explained, the deployment of the platform **621** may be enabled based on a relative position between the lid portion **620** and the base portion **610**. Similarly, the rotation of the platform **621** with respect to an upper side of the lid portion **620** about the platform hinge **628** may be enabled based on the relative position between the lid portion **620** and the base portion **610**. In an example, the hinge assembly **630** comprises a first locking member to lock a deployment of the platform **621** and the platform hinge **628** comprises a second locking member to lock a rotation of the platform **621**. Upon the angle between the lid portion **620** and the base portion **610** is to exceed a first threshold angle, a deployment of the platform **621** is unlocked by the first locking member. Then, once the platform **621** is fully deployed, a rotation of the platform **621** about the platform hinge **628** may be enabled by the second locking member. In an example, the rotation of the platform **621** about the platform hinge **628** is unlocked upon lid portion **620** is at an angle greater than a second threshold angle with respect to the base portion **610**, the second threshold angle being equal or greater than the first threshold angle.

(65) In an example, instead of preventing the platform **621** from moving and rotating based on the first threshold angle and the second threshold angle, a first threshold angle range for the deployment of the platform **621** and a second threshold angle range for the rotation of the platform **621** may be defined. In an example, the first threshold angle range comprises angles from 180 degrees to 270 degrees and the second threshold angle range comprises angles from 200 degrees to 250 degrees. However, in other examples, the first threshold angle range and the second threshold angle range may be the same, such as a range from 180 degrees to 270 degrees.

(66) In some other examples, the hinge assembly **630** may comprises a sensor to determine an angle between the lid portion **620** and the base portion **610** and a locking member to lock a movement of the platform **621** with respect to the lid portion **620** and a rotation of the platform **621** about the platform hinge **628**. Upon the determined angle is within a threshold angle range, the locking member is to unlock the movement of the platform **621**. Then, if the determined angle is lower than a locking angle within the threshold angle range, the locking member is to unlock the rotation of the platform **621**. In an example, the threshold angle range may be from 180 degrees to 270 degrees and the locking angle may be 250 degrees. Hence, if the determined angle is between 180 degrees and 250 degrees, the movement of the platform **421** and the rotation of the platform

421 about the hinge **427** may be enabled. If the determine angle is between 250 degrees and 270 degrees, the movement of the platform **421** is enabled but the locking member prevents the rotation of the platform **421** about the hinge **427**. In other examples, the computing device **600B** may comprise one of the sensors **326**, **326A** previously explained in FIGS. **3A** and **3C**, where the sensors **326**, **326A** may determine the relative position between the lid portion **620** and the base portion **610**.

(67) Referring now to FIG. **7A**, a computing device **700** comprising a stand feature **740** aligned with a surface **710B** of the base portion **710** is shown. The computing device **700** may correspond, for instance, with the computing devices **100**, **200**, **300**, **400**, **500A**, **500B**, **500C**, **600A**, and **600B** previously explained in reference to FIGS. **1A** to **6B**. The stand feature **740** is movable between a retracted position in which the stand feature **740** is aligned with the surface **710B** and a deployed position in which the stand feature **740** protrudes from the surface **710B**. In FIG. **7A**, the stand feature **740** is in the retracted position.

(68) In an example, the computing device **700** may use the stand feature **740** as a stand when the computing device **700** is to be used in a portrait orientation. In some examples, the computing device **700** may further comprise sensors to determine if the computing device **700** is in the portrait orientation, where the sensor is to issue a signal to a controller of the computing device **700** if the computing device **700** is determined to be in the portrait orientation. In response to the signal, the controller may rotate an image content to be displayed on the flexible display panel of the computing device **700**.

(69) Referring now to FIG. **7B**, the computing device **700** having the stand feature **740** in the deployed position is shown. In the deployed position, the stand feature **740** extends away from a recess **741** on the surface **710B**, i.e., protrudes from the surface **710B**. The computer device **700** further comprises an actuator **742** within the recess **741**, wherein the actuator **742** is to move the stand feature **740** between the retracted position (represented in FIG. **7A**) in which the stand feature **740** is aligned with the surface **710B** and the deployed position (represented in FIG. **7B**) in which the stand feature **740** extends from the recess **741** of the surface **710B**.

(70) In some examples, the position of the stand feature **740** may be controlled based on a position of a platform **721** of the computing device **700** with respect to the lid portion **720** of the computing device **700**. For instance, upon deploying the platform **721**, the stand feature **740** may protrude from the surface **710B** of the base portion **710**. In an example, the position of the stand feature **540** may be controlled based on a stand feature deployment signal issued by the controller of the computing device **700**. In response to the stand feature deployment signal, the actuator **742** is to move the stand feature **740** to the deployed position.

(71) Although the stand feature **740** of the computing device **700** of FIGS. **7A** and **7B** is located on the surface **710B** of the lid portion **710**, it should be understood that other alternative locations may be possible, such as a stand feature **740** on a lateral side of the base portion **710** or a stand feature **740** on a surface of the lid portion **720** of the computing device **700**.

(72) Referring now to FIG. **8**, a method **800** for deploying a platform of a lid portion of a computing device is shown. The computing device may correspond with any of with the computing devices **100**, **200**, **300**, **400**, **500A**, **500B**, **500C**, **600A**, **600B**, and **700** previously explained in reference to FIGS. **1A** to **7B**. At block **810**, method **800** comprises receiving a trigger event associated with a deployment of a platform of a lid portion of a computing device. As previously explained, the platform is covered by a flexible display panel disposed in the lid portion. At block **820**, method **800** comprises in response to the trigger event, deploying the platform from a retracted position to a deployed position through an aperture of the lid portion. In the deployed position, the platform extends away from an aperture of the lid portion. In an example, a first end of the flexible display panel of the computing device is attached to the lid portion and an interface region between the flexible display panel and an upper side of the lid portion is defined. As a result of the deployment (block **820**), the interface region is greater with respect to the interface region

before the deployment (in other words, the interface region is greater in the deployed position than in the retracted position).

(73) In some examples, when the computing device comprises a stand feature, method **800** may further comprise moving the stand feature to a deployed position in response to the trigger event. In order to deploy the stand feature, the computing device may comprise an actuator (for instance, the actuator **742** previously explained in FIG. 7B).

(74) In some other examples, method **800** may further comprise determining a deployment state of the platform with a sensor and displaying an image content on the flexible display panel based on the deployment state.

(75) Referring now to FIG. 9, a method **900** for unlocking a movement of a platform of a computing device upon an angle between the lid portion and the base portion is greater than a threshold angle is shown. At block, method **900** comprises receiving a trigger event associated with a deployment of a platform of a lid portion of a computing device. At block **920**, method **900** comprises deploying the platform from a retracted position to a deployed position through an aperture of the lid portion in response to the trigger event. Then, at block **930**, method **900** comprises determining an angle between the lid portion and the base portion. As previously explained, the computing device may comprise a sensor to determine a relative position between the lid portion and the base portion. At block **940**, method **900** comprises unlocking the movement of the platform upon the angle is greater than a threshold angle. In order to lock the movement of the platform, the computing device may comprise a locking member to lock the movement of the platform with respect to the aperture of the lid portion. In an example, method **900** may further comprise deploying a stand feature in response to a trigger event.

(76) In some examples, an end of the platform of the computing device comprises a hinge and a locking member to lock a rotation of the platform with respect to the lid portion and method **900** may further comprise unlocking the rotation about the hinge upon the angle between the lid portion and the base portion is greater than a second threshold angle.

(77) As explained above, examples of trigger events to trigger a deployment of a platform comprise receiving a trigger signal associated with a deployment of the platform, exceeding a threshold angle between the lid portion and the base portion, and actuating an actuator member located in the lid portion of the computing device.

(78) What has been described and illustrated herein are examples of the disclosure along with some variations. The terms, descriptions, and figures used herein are set forth by way of illustration only and are not meant as limitations. Many variations are possible within the scope of the disclosure, which is intended to be defined by the following claims (and their equivalents) in which all terms are meant in their broadest reasonable sense unless otherwise indicated.

Claims

1. A computing device comprising: a base portion; and a lid portion rotatably coupled to the base portion, the lid portion comprising: a platform; an aperture to receive the platform; and a flexible display panel disposed on the platform, wherein upon a deployment of the platform, an area of the flexible display panel on an upper side of the lid portion is to increase such that the area overlaps with the base portion.
2. The computing device of claim 1, wherein: a first edge of the flexible display panel is fixedly attached to a first end on the upper side of the lid portion, the flexible display panel and a lower side of the platform overlap in an overlap area, and upon the deployment of the platform, the overlap area is to decrease with respect to the overlap area before the deployment of the platform.
3. The computing device of claim 1, wherein the lid portion includes a locking member to lock a position of the platform with respect to the lid portion, and in response to a trigger event, the locking member is to enable the deployment of the platform.

4. The computing device of claim 1, further comprising a stand feature aligned with a surface of the base portion, wherein upon the deployment of the platform, the stand feature is to protrude from the surface of the base portion.
 5. The computing device of claim 4, wherein: the lid portion further comprises a deployment member to move the platform to a deployed position, and in response to a trigger event, the deployment member is to move the platform to the deployed position.
 6. The computing device of claim 4, wherein the lid portion further comprises a sensor to determine a relative position between the upper side of the lid portion and an upper side of the base portion, wherein the sensor is to issue a trigger event upon the upper side of the lid portion is substantially parallel to the upper side of the base portion.
 7. A method comprising: receiving a trigger event associated with a deployment of a platform of a lid portion of a computing device rotatably coupled to a base portion of the computing device, the platform being covered by a flexible display panel disposed in the lid portion; and in response to the trigger event, deploying the platform from a retracted position to a deployed position through an aperture of the lid portion, wherein a first end of the flexible display panel is attached to the lid portion, wherein an interface region between the flexible display panel and an upper side of the lid portion is greater in the deployed position than in the retracted position.
 8. The method of claim 7, further comprising: determining a deployment state of the platform associated with the interface region; and displaying an image content on the flexible display panel based on the deployment state of the platform.
 9. The method of claim 7, wherein the lid portion further comprises a locking member to lock a movement of the platform with respect to the aperture, the method further comprising: determining an angle between the lid portion and the base portion; and upon the angle is greater than a threshold angle, unlocking the movement of the platform.
 10. The method of claim 9, wherein the trigger event comprises: receiving a trigger signal associated with a deployment of the platform; exceeding the threshold angle; and actuating an actuator member located in the lid portion.
 11. A computing device comprising: a base portion; a lid portion rotatably coupled to the base portion, the lid portion comprising: a platform; an aperture to receive the platform; a flexible display panel to cover an upper side of the lid portion and a lower side of the platform, wherein a first end of the flexible display panel on the upper side is attached to the lid portion; a track along a lateral side of the lid portion; and a protruding element on a lateral side of the platform; wherein the protruding element is movable along the track between a first position and a second position, wherein upon a movement of the protruding element from the first position towards the second position, the flexible display panel is to extend away from the aperture.
 12. The computing device of claim 11, wherein an end of the platform comprises a hinge, wherein upon the protruding element is in the second position, the platform is to rotate about the hinge with respect to the lid portion.
 13. The computing device of claim 12, wherein the platform comprises a torsion element fixedly attached to a second end of the flexible display panel such that the torsion element is to tighten the flexible display panel on the upper side of the lid portion.
 14. The computing device of claim 11, wherein the lid portion is rotatably coupled to the base portion via a hinge assembly, the hinge assembly comprising: a sensor to determine an angle between the lid portion and the base portion, and a locking member to lock a movement of the platform with respect to the lid portion, wherein upon the angle is within a threshold angle range, the locking member is to unlock the movement of the platform.
 15. The computing device of claim 14, wherein the locking member is further to lock a rotation of the lid portion with respect to the base portion, wherein if the angle is lower than a locking angle within the threshold angle range, the locking member is to unlock the rotation.
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